

Business Insights Report (EDA)

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on a marketing campaign dataset to understand customer purchasing behavior, analyze campaign performance, and identify marketing channels that provide better return on investment (ROI).

Data Cleaning & Preparation

The dataset was cleaned by handling missing values, removing duplicates, and creating a new feature called "Total Spending" by combining customer spending across multiple product categories.

Tools used:

- Python
- Pandas
- Matplotlib
- Seaborn

Key Findings

Customer Spending Behavior

Analysis shows that customers with higher income spend significantly more on products. Income is positively correlated with total spending.

Marketing Channel Performance

Store purchases generated the highest number of purchases, followed by web purchases. Catalog purchases had the lowest performance.

Campaign Performance

Campaigns 4 and 5 received the highest acceptance rates from customers, indicating stronger engagement and campaign effectiveness. Campaign 2 showed the lowest acceptance rate.

Product Purchase Relationships

Customers purchasing one category of products often purchased products from other categories as well, indicating strong cross-selling opportunities.

ROI Recommendations

- Increase investment in store and digital marketing channels.
- Focus future campaigns on strategies similar to Campaigns 4 and 5.
- Reduce spending on catalog marketing and low-performing campaigns.
- Target high-income customer segments for premium product promotions.
- Improve website conversion strategies to convert visitors into buyers.

Conclusion

The EDA process helped identify customer trends, effective marketing channels, and successful campaigns. The findings suggest that the company can improve ROI by focusing more on high-performing channels and customer segments while optimizing underperforming campaigns.